

CSCI 5880

Interactive Machine Learning for Customizable and Expressive Interfaces

FORMAT	CATEGORY	CREDITS
On-Campus	Elective	None
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective (On-Campus)	Machine Learning Engineer (45.3 / 100)	36.1 / 100

36

CURRENT AVG SCORE
across 6 job roles

82

POTENTIAL SCORE
with recommended additions

+58

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

Teaches machine learning application for human-computer interfaces, processing video and accelerometer streams for interactive systems.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
Machine Learning Engineer	45.3	82.5	+37.3	HIGH
MLOps Engineer	40.0	92.3	+52.3	HIGH
AI Engineer	38.2	89.1	+50.9	HIGH
ML Engineer	35.3	70.7	+35.4	HIGH
Applied Scientist	30.8	73.0	+42.2	HIGH
Data Scientist	27.1	84.6	+57.5	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

Deep Learning 1 of 6 roles

cnn

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1 **LOW** **Optional: Deep Learning** — would broaden appeal to one additional role. Candidate terms: cnn.

GAP KEYWORDS BY ROLE

Machine Learning Engineer

45.3 → 82.5 (+37.3 pts)

data web sets tools exploratory design development science

MLOps Engineer

40.0 → 92.3 (+52.3 pts)

detection data model text image tools models development

AI Engineer

38.2 → 89.1 (+50.9 pts)

perception following data robotics models build technical model

ML Engineer

35.3 → 70.7 (+35.4 pts)

cnn data models model world build problems product

Applied Scientist

30.8 → 73.0 (+42.2 pts)

research problems models technical data industry design domains

Data Scientist

27.1 → 84.6 (+57.5 pts)

data sets product mitigation different behaviors social build

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.